

Method	Trainable	Total	% Trainable	Throughput	HM
	(M)	(M)		(img/s)	(%)
CoOp	<u>0.01</u>	149.62	<u>0.007</u>	1485	71.66
CoCoOp	0.36	149.98	0.24	983	75.83
MaPLe	3.55	153.17	2.37	1320	78.55
MMRL	4.21	153.83	2.81	1265	<u>80.44</u>
MMRL++	0.78	150.40	0.52	<u>1432</u>	81.08

Table 1: Parameter-efficiency comparison. MMRL++ achieves the best harmonic mean (81.08%) while training only 0.52% of the network parameters — $5.4 \times$ fewer than MMRL and $4.6 \times$ fewer than MaPLe — and the second-highest inference throughput. Throughput measured at batch size 32 on a single A100 GPU.